

Lumina Renewables S.p.A.

Monte Carlo DCF Valuation • A plain-language walk-through • 50,000 iterations

Lumina Renewables is a mid-market Italian solar & wind operator. This report values its equity, but instead of producing a single number it produces a **range of possible values and the probability of each**. Because the business is capital-intensive and exposed to energy taxation, two inputs dominate the result: the discount rate (**WACC**) and the **tax rate**.

A normal DCF gives one answer — say €14.92 per share — built on six assumptions that are really just educated guesses. A Monte Carlo replaces each guess with a **probability distribution**, then re-runs the valuation 50,000 times, each time drawing fresh random values. The output is the full distribution of what the share could be worth.

How to read this report

- **The inputs.** What a distribution is, and what “Min / Mode / Max” and “Mean ± SD” mean.
- **The result.** How to read the histogram, percentiles, median vs mean, and probabilities.
- **What matters.** Which assumptions move the valuation most (the tornado).
- **The odds.** The cumulative curve — reading the probability of any outcome directly.
- **The journey.** Simulated price paths over time — the “spaghetti” fan chart.
- **Limits & glossary.** What the method cannot do, plus definitions of every term used.

BASE CASE / SHARE

€14.92

SIMULATED MEDIAN (P50)

€12.46

AVERAGE (MEAN)

€13.23

DOWNSIDE (P5)

€5.35

UPSIDE (P95)

€23.63

P(VALUE > MARKET €11.50)

57%

The one thing to remember: the single-point base case of €14.92 sits **above** the simulated median of €12.46. The plain DCF flatters the valuation because discounting reacts non-linearly to the WACC. The realistic central estimate is the **median**, inside a wide €5.35–€23.63 band.

SECTION 1

The inputs — turning guesses into distributions

Every valuation rests on assumptions. A standard model uses one value per assumption (“growth is 9%”). But nobody knows growth will be exactly 9% — it could be 5% or 12%. A **probability distribution** captures that honestly: it describes the whole range a number could take and how likely each value is. We use two shapes.

Triangular — defined by Min / Mode / Max

The simplest way to express an estimate with a range. You give three numbers: the lowest plausible value (**Min**), the single most likely value (**Mode**), and the highest plausible value (**Max**). Likelihood rises to the mode and falls after it — literally a triangle.

Example (tax rate 24% / 26% / 35%): it will not fall below the 24% IRES base, most likely sits at 26%, but could climb to 35% in a windfall-tax scenario. The wide gap on the high side is a deliberate **upward skew** — we think a tax increase is more likely than a tax cut.

Normal — defined by Mean ± SD

The familiar bell curve: symmetric around a central value (the **Mean**). The **SD** (standard deviation) sets the width — about 68% of outcomes land within one SD of the mean, 95% within two. We use it for WACC, which clusters symmetrically around a best estimate (8.5%, give or take 1.1%).

Correlation (+0.35). Some inputs move together: when revenue grows faster, margins tend to rise too (scale economies), so we link revenue growth and EBITDA margin instead of treating them independently. Ignoring this understates how extreme good and bad scenarios can get.

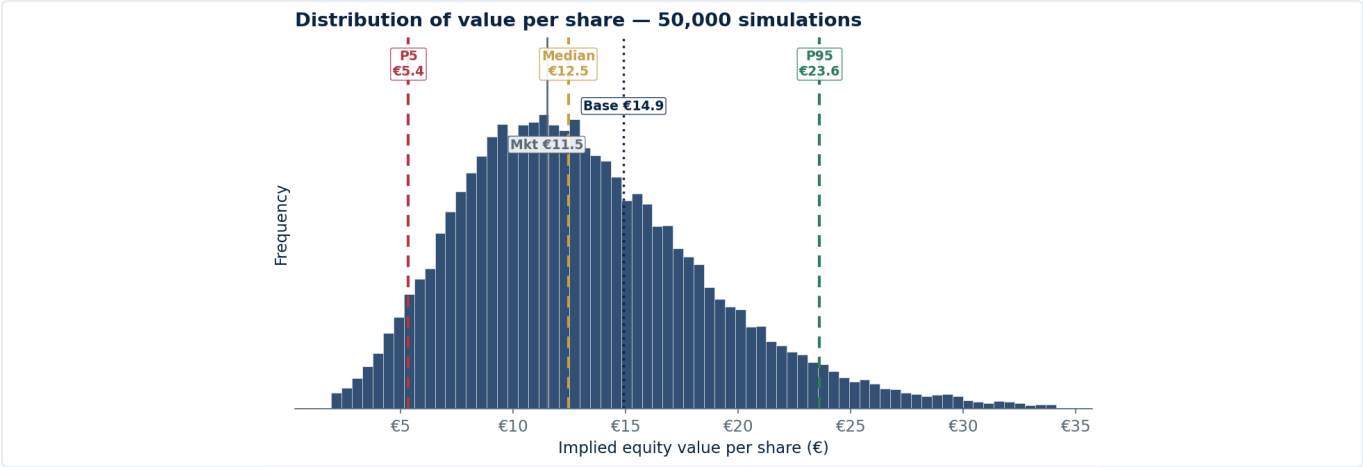
Variable	Distribution	Values	Read as
Revenue growth (p.a.)	Triangular	4% / 9% / 14%	Min / Mode / Max
EBITDA margin	Triangular	36% / 42% / 46%	Min / Mode / Max
WACC	Normal	8.5% ± 1.1%	Mean ± SD
Tax rate	Triangular	24% / 26% / 35%	Min / Mode / Max
Terminal growth	Triangular	1.0% / 2.0% / 2.75%	Min / Mode / Max
CapEx (% of revenue)	Triangular	12% / 16% / 22%	Min / Mode / Max

Fixed (non-random) parameters: base revenue €145m, D&A 13% of revenue, ΔNWC 3% of incremental revenue, net debt €185m, 25.0m shares. A 1.5% minimum WACC–terminal-growth spread is enforced so extreme draws cannot produce an infinite terminal value.

SECTION 2

The result — a distribution, not a single number

Running the model 50,000 times produces 50,000 possible share values. The histogram below groups them into bars: the taller the bar, the more often that value occurred. The shape is **right-skewed** — a long tail of high outcomes on the right.



Percentiles. Sort all outcomes from low to high. A percentile is a cut-off point: **P50** (the median) is the middle value — half the outcomes fall below it, half above. **P5** means only 5% of outcomes are worse; **P95** means only 5% are better. The **P5–P95 band** (€5.35 –€23.63) is therefore a 90% confidence range.

Median vs mean. The mean (€13.23) is the simple average; the median (€12.46) is the middle outcome. They differ here because the right tail drags the average up. The **median is the more representative “typical” value**.

Level	€/sh	Meaning
P5 — downside	€5.35	only 5% of outcomes are worse
P25	€9.18	lower quartile
P50 — median	€12.46	the typical outcome
P75	€16.42	upper quartile
P95 — upside	€23.63	only 5% of outcomes are better

What the probabilities mean

A probability read-out is simply the **share of the 50,000 simulated worlds** in which a condition held. It turns the distribution into a direct risk statement:

- Value above the €11.50 market price: **57%** — more likely than not, but far from certain.
- Value below the base case (€14.92): **67%** — the base case is optimistic.
- A severe outcome below €8.00: **17%** — the realistic tail risk.

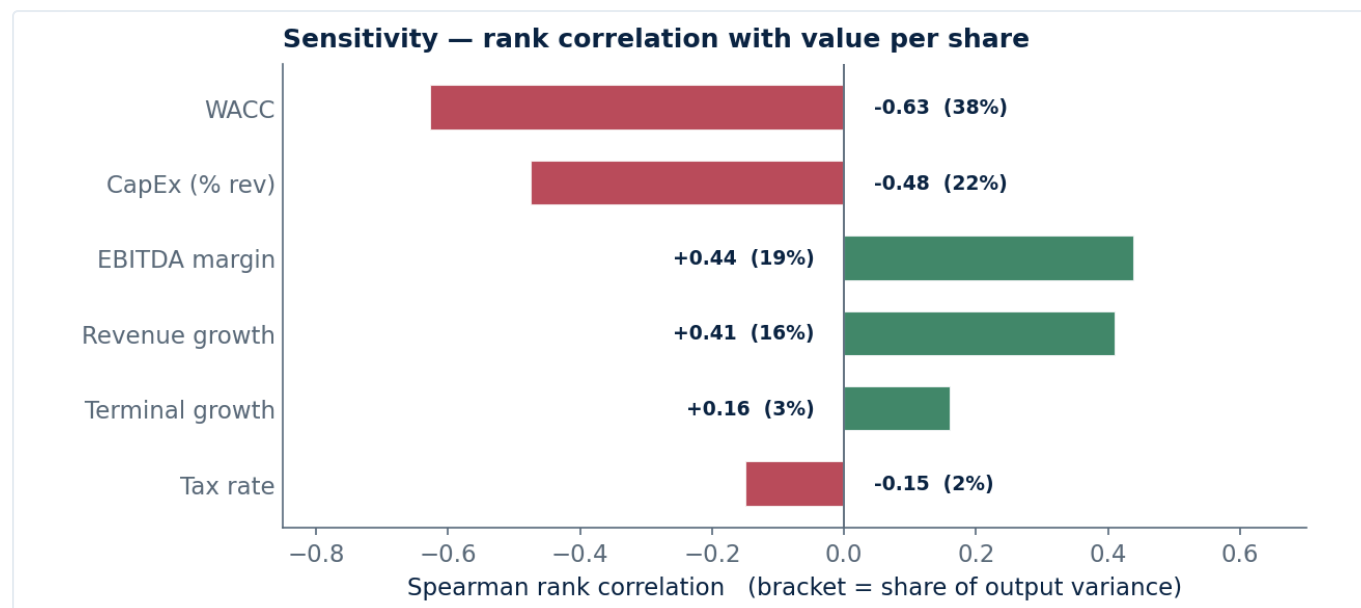
SECTION 3

What matters — which assumptions move the value

Not all assumptions carry equal weight. The tornado chart ranks them by how strongly each one drives the final value. We measure this with **rank correlation**: a number between -1 and $+1$.

- Close to **+1**: when this input goes up, the valuation reliably goes up (and it moves it a lot).
- Close to **-1**: when this input goes up, the valuation reliably goes **down**.
- Close to **0**: this input barely affects the result.

The bracketed figure is each input's approximate **share of total variance** — how much of the overall uncertainty it explains. Bars are sorted widest-to-narrowest, giving the “tornado” shape; the top bar is the biggest lever.



Driver	Rank corr.	Share of variance	Effect
WACC	-0.63	38%	decreases value
CapEx (% of revenue)	-0.48	22%	decreases value
EBITDA margin	+0.44	19%	increases value
Revenue growth	+0.41	16%	increases value
Terminal growth	+0.16	3%	increases value
Tax rate	-0.15	2%	decreases value

Reading it: WACC is by far the single largest swing factor, followed by capital intensity (CapEx) and EBITDA margin. The tax rate genuinely matters — a move toward the 35% windfall scenario erodes value — but it is a **second-order** driver here, not the dominant one. Practically: spend diligence effort narrowing the WACC and CapEx estimates first, because that is where most of the uncertainty lives.

SECTION 4

The odds — reading the cumulative curve

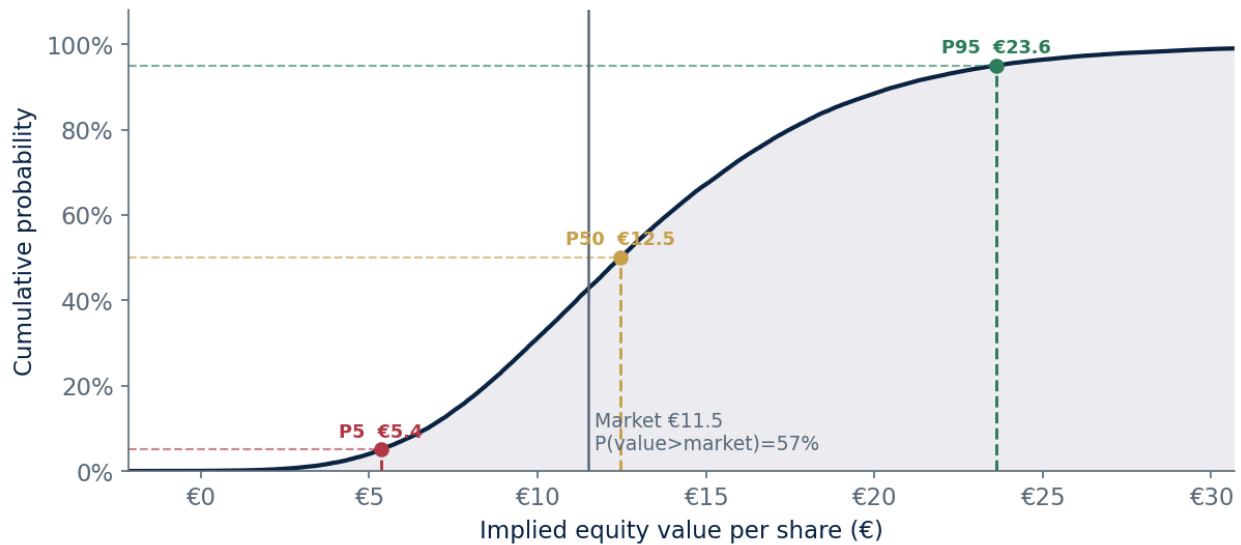
The S-curve is the histogram re-expressed as cumulative probability. For any value on the horizontal axis, the height of the curve tells you the probability the outcome lands **at or below** that value. It rises from 0% (nothing is below the minimum) to 100% (everything is below the maximum).

Think of it as the running total of the histogram: at each point on the axis the curve adds up the probability of every outcome at or below it. That makes it the most decision-friendly view of the whole simulation — rather than eyeballing the height of bars, you read a probability straight off the vertical axis.

Quick reads. Half of all outcomes fall below the median (€12.46); the middle 50% sit between €9.18 (P25) and €16.42 (P75); and only one outcome in ten clears €20.63 (P90), while one in ten falls below €6.73 (P10).

Slope matters too. Where the curve is steep, outcomes are tightly bunched; where it flattens, they are sparse and spread out. The long, slow climb through the upper tail is the visual signature of the right-skew — the upside is more open-ended than the downside.

Cumulative probability (S-curve) of value per share



Two ways to read it

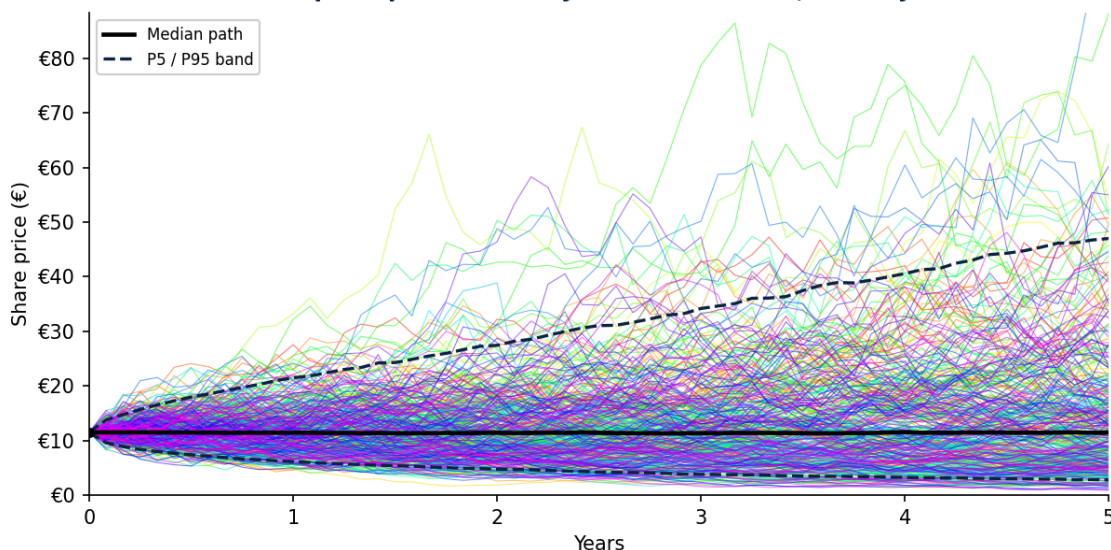
- **Read vertically** at a value you care about to get its probability. At the €11.50 market price the curve is at roughly 43%, so about 57% of outcomes sit above the market — i.e. the probability the stock is undervalued.
- **Read horizontally** from a probability to find the matching value. The 50% line meets the curve at the median (€12.46); the 5% and 95% lines give the downside and upside (€5.35 and €23.63).

SECTION 5

The journey — simulating the price path over time

Every chart so far collapsed the five-year forecast into a single value. This one keeps the **time dimension**: each thin line is one simulated future path of the share price, and all 10,000 of them start from today's €11.50. It answers a different question — not “what is it worth?” but “how might the price *travel*?”

Simulated share-price paths over 5 years — 400 of 10,000 trajectories



The engine is **Geometric Brownian Motion**, the standard model for asset prices. Each month the price is nudged by a random shock: a steady **drift** (7% per year, the average pull) plus **volatility** (38% per year, the size of the randomness). Because those random monthly moves **compound**, the spread of outcomes widens with the square root of time — producing the characteristic **cone**. The black line is the median path; 90% of trajectories stay inside the dashed P5–P95 band.

What the cone says

- After five years the price ranges from about €2.79 (P5) to €47.01 (P95), median €11.46.
- Probability the price more than doubles: **21%**; probability it ends below today's level: **50%**.
- The median barely moves from €11.50: at this volatility the random downside drag offsets the drift — high volatility quietly penalises the typical path even when the average return is positive.

This path simulation is deliberately separate from the DCF valuation: that one re-valued the company 50,000 ways *today*; this one evolves the market price *through time* under an assumed drift and volatility. Together they cover both the level and the trajectory of value.

SECTION 6

How to use this — and what it cannot do

- **Probabilities, not prophecies.** The output is a distribution; the realised value will be a single point inside it.
- **Garbage in, garbage out.** Results are only as good as the input ranges. These ranges are illustrative judgement, not live market data.
- **Known unknowns only.** The model prices the risks we defined. It cannot capture true black swans — regulatory shocks or demand collapse outside the modelled ranges.
- **Convexity caution.** Low-WACC draws inflate terminal value disproportionately; the 1.5% spread guardrail limits, but does not remove, this effect.

Glossary

Distribution — the full set of values a number can take, with the likelihood of each, instead of one fixed figure.

Triangular (Min / Mode / Max) — an estimate given as lowest, most-likely, and highest plausible values.

Normal (Mean $\pm$ SD) — a symmetric bell curve; SD (standard deviation) sets its width.

Percentile (P5, P50, P95) — a cut-off in the ranked outcomes; PX means X% of outcomes fall below it.

Median (P50) — the middle outcome; half above, half below. **Mean** — the simple average.

Correlation — the tendency of two inputs to move together (+) or oppositely (–).

WACC — weighted average cost of capital; the rate used to discount future cash flows to today.

Terminal value — the value of all cash flows beyond the explicit 5-year forecast.

Rank correlation / tornado — a measure (–1 to +1) of how strongly each input drives the output.

Unlevered FCF — cash a business generates before financing; what a DCF discounts.

Geometric Brownian Motion (GBM) — the standard model for a price that drifts upward on average while taking random steps each period.

Drift & volatility — in GBM, the average yearly pull (drift) and the size of the random shocks (volatility).